

5(a)	(α -)glucose and fructose ; R β -glucose I α - or β - in front of fructose	1
5(b)	1 sucrose, diffuses / moves down the concentration gradient, from mesophyll cell, through plasmodesmata / in the symplast pathway, to the phloem parenchyma cell ; or sucrose, diffuses / moves down the concentration gradient, from phloem parenchyma cell, through plasmodesmata / in the symplast pathway, to the companion cell ; <i>and any four from mp2 to mp6:</i> <i>phloem parenchyma cell to companion cell</i> 2 sucrose moves out of phloem parenchyma cell to, cell wall / apoplast A diffuses out (because of part (c)) or diffusion / movement, through, apoplast / cell wall, of (phloem) parenchyma cell to (apoplast / cell wall of) companion cell ; 3 <i>ref. to</i> proton pump / ATP required, for movement of protons out of companion cell (into companion cell wall) ; A active transport of protons out of companion cell 4 sucrose cotransported with, hydrogen ions / protons / H^+, into companion cell ; A sucrose and, H^+/protons, enter companion cell through cotransporter (protein) 5 movement of sucrose against the concentration gradient (into companion cell) ; 6 movement of, H^+/protons, by facilitated diffusion ; 7 sucrose, diffuses / moves down the concentration gradient, (from companion cell) into phloem sieve tube element through plasmodesmata ; 8 AVP ; e.g. movement out of sucrose from (phloem) parenchyma cell to apoplast by facilitated diffusion	5

Question	Answer	Marks
5(c)(i)	any three from: 1 (SWEET is a) transmembrane / integral / intrinsic / carrier (protein) ; bod channel protein A described 2 protein extends, above and below / beyond, phospholipid bilayer / membrane ; 3 single polypeptide ; 4 seven alpha helices (within cell (surface) membrane) ; A protein is composed of alpha helices A (secondary) structure of protein is alpha helix R protein has alpha helices and beta pleated sheets 5 helices are, parallel / AW ; ecf from MP3 – polypeptides are, parallel / AW 6 (helices form a) hydrophilic, pathway / pore / channel / AW (for sucrose) ; 7 (SWEET is a) globular protein ; 8 AVP ; e.g. random coils	3
5(c)(ii)	any two from: sucrose is, a polar molecule / (water) soluble / hydrophilic / is not lipid soluble ; will not pass through, hydrophobic core of membrane / fatty acid tails of phospholipids / AW ; SWEET protein allows a hydrophilic route (through membrane) ; SWEET protein allows facilitated diffusion ;	2